

# A Hybrid Machine Learning Pipeline for Differential Diagnosis Using Indirect Ontology-Aligned Evidence Mapping

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## Abstract

Clinical diagnosis is rarely straightforward and deterministic. A large portion of the information that clinicians rely on is ambiguous or insufficient, patients frequently describe their conditions in everyday language, and symptoms often overlap across diseases. Although diagnosis is typically reduced to a single-label classification task by machine learning systems, this abstraction ignores the ambiguity that is essential to true clinical reasoning.

In this work, we present a hybrid diagnostic pipeline that uses a lightweight yet semantically enriched approach to connect structured medical evidence with informal patient descriptions. `mapping_E_inferred.json`, a custom alignment resource, links interpretable symptom ontology terms to anonymised evidence identifiers from the DDXPlus dataset. The fuzzy text-to-evidence extraction procedure is strengthened by these inferred tokens. An XGBoost classifier receives the resulting evidence vectors and outputs a complete probability distribution over diagnoses. The system outperforms more sophisticated reinforcement-learning diagnostic agents despite its simplicity, achieving 99.72% accuracy on DDXPlus.

## 1 Introduction

Making a diagnosis is rarely a simple or straightforward process. The symptoms that patients report do not neatly map onto the vocabulary that medical datasets or diagnostic engines expect, and doctors frequently receive incomplete, subjective, or inconsistent information

in actual clinical settings. Anxiety, asthma, a respiratory infection, anaemia, or even heart failure could all be indicated by a single complaint like "I feel short of breath." The diagnostic process is typically framed as a search through several plausible explanations rather than a quest for a single, definitive label because multiple conditions can present with similar clusters of symptoms. A *differential diagnosis*, which is a prioritised list of conditions that could plausibly explain the patient's presentation, is created by clinicians. As new information becomes available, this list is updated.

However, diagnosis has historically been reduced by machine learning systems to a one-shot prediction problem, in which a model is fed a fixed feature vector and required to produce a single "correct" label. Although this framing is helpful from an algorithmic standpoint, it eliminates the ambiguity and uncertainty that characterise true clinical judgement. Additionally, it ignores the fact that diagnostic reasoning frequently involves iterations. A physician does more than just observe symptoms; they also interpret them, ask for more information, take into account uncommon possibilities, and revise their opinions as new information becomes available.

Current diagnostic datasets make an effort to close this gap. The DDXPlus dataset is unique among them since it explicitly offers ranked differential diagnoses for 1.3 million fictitious patient encounters. DDXPlus is created by an artificial diagnostician intended to mimic structured medical reasoning, in contrast to standard EHR datasets that document diagnoses retroactively and frequently inconsistently. A ranked list of potential diseases, demographics, antecedents, and binary and multichoice evidence fields are all included in each encounter. Despite being artificial, this structure captures significant elements of clinical uncertainty, which makes it appropriate for training probabilistic reasoning models.

But DDXPlus presents new difficulties. Anonymised evidence identifiers, like E\_57 or E\_162, are used in the dataset to express symptoms; these identifiers are distinct from the ontology that the underlying diagnostic engine uses. This indicates that there isn't a clear, machine-readable connection between the symptom concepts found in the knowledge base and the dataset features. This disconnect becomes a major barrier for real-world use, where patients describe symptoms in free text. Text such as "I get breathless when walking" cannot be meaningfully interpreted by a model unless it can be mapped to the appropriate evidence ID in the dataset. A system that matches structured, anonymised dataset features with natural language descriptions is needed to close that gap.

This challenge was the driving force behind the work presented in this paper. Our objective was to create a system that could translate unstructured, user-written symptom descriptions into structured evidence vectors that could be used for diagnostic prediction. We pursued a lighter and more comprehensible pipeline rather than depending exclusively on deep neural architectures or large language models. A custom semantic resource we created, `mapping_E_inferred.json`, is used to implement an indirect ontology-aligned evidence mapping strategy at its core. Using co-occurrence patterns, question text similarity, and pathology associations, this file connects anonymised evidence fields to symptom ontology concepts.

Fuzzy text matching becomes much more dependable once this semantic alignment is estab-

lished. Although none of these exact strings appear in the dataset, terms like "breathlessness," "short of breath," "air hunger," and "dyspnoea" all suggest the same underlying symptom. The system gains a more human-like comprehension of symptom descriptions while maintaining computational lightness and ease of interpretation by adding ontology tokens to the phrase bank.

This work suggests a hybrid pipeline that combines a strong gradient-boosted classifier with ontology-enriched fuzzy evidence extraction. The pipeline concentrates on accurate static prediction when all available evidence is known, rather than trying to mimic the sequential reasoning of clinicians or reinforcement-learning agents. The system outperforms more sophisticated RL-based diagnostic models in the full-information setting, achieving 99.72% accuracy on the DDXPlus test split despite its simplicity. The alignment resource’s design, the hybrid pipeline’s architecture, comparisons with current diagnostic systems, and possible ways to expand this work to actual clinical settings are all covered in the remaining sections of this paper.

## 2 Literature Review

The DDXPlus dataset is now one of the most popular synthetic diagnostic benchmarks, and research on automated medical diagnosis has grown significantly in recent years. Numerous studies examine various modelling techniques, such as ontology-driven reasoning, supervised learning, and reinforcement learning. The most pertinent studies that directly use or expand DDXPlus are compiled below.

- **DDXPlus: A New Dataset for Automatic Medical Diagnosis (Tchango et al., 2022)** The DDXPlus dataset was first presented in this seminal paper, which also described its motivation, structure, and evaluation procedures. It suggested using DDXPlus as a testbed for automatic differential diagnosis and multi-hypothesis reasoning. This dataset serves as the foundation for our entire project, and many of our techniques were developed by examining its features.
- **Towards Trustworthy Automatic Diagnosis Systems (CASANDE) (2022)** CASANDE trains a reinforcement-learning agent that can ask interactive diagnostic questions using the DDXPlus simulator. The study showed that in situations where information is lacking, sequential evidence acquisition may perform better than static classifiers. In this project, we show instances where static full-evidence prediction is better by comparing the predictive accuracy of our model with CASANDE.
- **AARLC and BASD Diagnostic RL Baselines (2022–2023)** These companion papers introduced RL baselines for DDXPlus such as AARLC and BASD. They looked into information-seeking behaviours, policy optimisation, and reward shaping. These studies supported our choice of a supervised framework by demonstrating that while RL agents can be useful in dialogue settings, they frequently yield lower top-1 accuracy when compared to supervised models when all the evidence is known.

- **DDXPlus Official GitHub Repository** RL baselines for DDXPlus, such as AARLC and BASD, were introduced in these companion papers. They investigated reward shaping, policy optimisation, and information-seeking behaviours. These studies showed that although RL agents can be helpful in dialogue settings, they often produce lower top-1 accuracy when compared to supervised models when all the evidence is known, which validated our decision to use a supervised framework.
- **Robustness of Diagnostic Models Under Noise (2023)** The robustness of models trained on DDXPlus in the face of noisy, incomplete, or corrupted evidence was examined in this work. The results demonstrated that under input noise, simple classifiers drastically deteriorate. In order to increase stability under a variety of user inputs, we included ontology-aligned fuzzy evidence mapping.
- **Adaptive Question Selection for Diagnosis (2023)** This study used DDXPlus to assess RL-based adaptive questioning techniques. It demonstrated how choosing the right questions effectively lowers uncertainty. These concepts affected our comprehension of sequential versus static diagnostic reasoning, even though our system does not engage in interactive questioning.
- **Benchmarking Classical and Deep Models on DDXPlus (2023)** RL-based adaptive questioning strategies were evaluated in this study using DDXPlus. It illustrated how selecting appropriate questions reduces uncertainty. Despite the fact that our system does not use interactive questioning, these ideas had an impact on our understanding of sequential versus static diagnostic reasoning.
- **Ontology-Aware Medical Reasoning Models (2023–2024)** These studies investigated the integration of DDXPlus with medical ontologies like SNOMED CT and UMLS. They showed enhanced comprehension of the evidence and interpretability. Our ontology-aligned evidence-to-text mapping method was motivated by their findings.
- **Synthetic Patient Simulation in Diagnostic AI (2024)** This line of research evaluated diagnostic agents in controlled environments using simulators similar to DDXPlus. The outcomes encouraged us to rely on synthetic encounters for safe experimentation and reaffirmed the significance of synthetic data in the development of early-stage diagnostic systems.
- **Transformer-Based Differential Diagnosis Models (2024–2025)** This field of study used simulators akin to DDXPlus to assess diagnostic agents in controlled settings. The results confirmed the importance of synthetic data in the development of early-stage diagnostic systems and encouraged us to rely on synthetic encounters for safe experimentation.

### 3 Methodology

In order to be applicable to any structured diagnostic dataset that includes clinical evidence, demographic characteristics, and disease labels, our methodology is purposefully general.

The pipeline consists of three main steps: (1) interpreting text provided by the patient; (2) converting the interpreted data into a numerical feature representation; and (3) using a probabilistic classifier to predict a complete differential diagnosis. Because each stage is modular, the system can adjust to different medical domains without changing the architecture as a whole.

### 3.1 Stage 1: Interpreting Patient Descriptions

The system must interpret text that seldom matches the controlled vocabulary used in structured datasets because patients usually describe their symptoms in free-form language. Let  $T$  denote a patient’s text input. After basic normalization and tokenization, our goal is to assign  $T$  to a subset of evidence indicators from a finite vocabulary  $\mathcal{E} = \{e_1, e_2, \dots, e_K\}$ .

Each evidence item  $e_i$  is associated with a phrase bank:

$$\mathcal{P}_i = \{p_{i1}, p_{i2}, \dots, p_{im}\},$$

where  $p_{ij}$  could be automatically generated synonyms, ontology terms, answer choices, or question text. We calculate three similarity signals with the input text  $T$  for any phrase  $p$ :

**1. Substring similarity:**

$$s_{\text{sub}}(T, p) = \frac{|T \cap p|}{|p|},$$

where  $T \cap p$  counts overlapping character substrings.

**2. Token overlap:**

$$s_{\text{tok}}(T, p) = \frac{|\text{Tokens}(T) \cap \text{Tokens}(p)|}{|\text{Tokens}(p)|}.$$

**3. Fuzzy edit-distance score:**

$$s_{\text{fuzz}}(T, p) = 1 - \frac{d_{\text{lev}}(T, p)}{\max(|T|, |p|)},$$

where  $d_{\text{lev}}$  is the Levenshtein distance.

We combine these signals into a unified matching score:

$$S(T, p) = \alpha s_{\text{sub}}(T, p) + \beta s_{\text{tok}}(T, p) + \gamma s_{\text{fuzz}}(T, p),$$

where  $\alpha, \beta, \gamma$  are non-negative weights satisfying  $\alpha + \beta + \gamma = 1$ .

The relevance of a patient text  $T$  to an evidence indicator  $e_i$  is then:

$$R(T, e_i) = \max_{p \in \mathcal{P}_i} S(T, p).$$

Evidence indicators exceeding a similarity threshold  $\tau$  are selected:

$$\hat{\mathcal{E}}(T) = \{e_i : R(T, e_i) \geq \tau\}.$$

Because the phrase banks and thresholds can be modified for any diagnostic dataset without changing the fundamental technique, this process is dataset-agnostic.

### 3.2 Stage 2: Structured Feature Construction

The selected evidence set  $\hat{\mathcal{E}}(T)$  must be converted into a numerical representation that is structured and appropriate for machine learning models. Assume that there are  $v_i$  possible categorical values for each evidence indicator (binary indicators are the special case where  $v_i = 2$ ).

We construct a feature vector  $x \in \mathbb{R}^d$  with components:

$$x = [x_{\text{evidence}}, x_{\text{demo}}, x_{\text{antecedent}}, x_{\text{prior}}].$$

**Evidence Encoding.** Each evidence indicator  $e_i$  is encoded using one-hot or multi-hot encoding:

$$x_{\text{evidence}}^{(i)}(v) = \begin{cases} 1 & \text{if value } v \text{ is detected,} \\ 0 & \text{otherwise.} \end{cases}$$

**Demographic Features.** Demographic attributes such as age, sex, or BMI are encoded generically:

$$x_{\text{demo}} = [\text{age}, \mathbb{1}_{\text{sex=male}}, \mathbb{1}_{\text{sex=female}}, \dots].$$

**Antecedent Encoding.** Antecedents (risk factors) are represented by:

$$x_{\text{antecedent}}^{(j)} = \begin{cases} 1 & \text{if antecedent } a_j \text{ is present,} \\ 0 & \text{otherwise.} \end{cases}$$

**Optional Prior Features.** Value frequencies or disease prevalence are included in some datasets. These can be included as:

$$x_{\text{prior}}^{(k)} = \log(\text{freq}(f_k) + 1).$$

Because it does not presuppose any particular format for evidence, this feature construction process is purposefully generic, making it appropriate for diagnostic datasets with arbitrary categorical structures.

### 3.3 Stage 3: Probabilistic Diagnostic Prediction

Let the set of diseases be  $\mathcal{Y} = \{1, 2, \dots, C\}$ . The goal is to predict a probability distribution:

$$P(y = c \mid x), \quad c \in \mathcal{Y}.$$

The XGBoost classifier, which maximises the softmax cross-entropy (multiclass logistic loss), is employed in our implementation:

$$\mathcal{L} = - \sum_{i=1}^N \log \frac{\exp(f_{y_i}(x_i))}{\sum_{c=1}^C \exp(f_c(x_i))}.$$

The model prediction for each class is:

$$P(y = c \mid x) = \frac{\exp(f_c(x))}{\sum_{k=1}^C \exp(f_k(x))}.$$

XGBoost computes tree-based functions:

$$f_c(x) = \sum_{t=1}^T f_{t,c}(x),$$

where  $f_{t,c}$  is the output of tree  $t$  for class  $c$ .

### 3.4 Ranking the Differential Diagnosis

Instead of returning a single predicted disease, we generate a ranked list:

$$\hat{y}_1, \hat{y}_2, \dots, \hat{y}_C$$

such that

$$P(y = \hat{y}_1 \mid x) \geq P(y = \hat{y}_2 \mid x) \geq \dots \geq P(y = \hat{y}_C \mid x).$$

Clinical differential diagnosis, where several options must be assessed in order of likelihood, naturally corresponds with this ranking.

### 3.5 Methodological Principles

Four design principles influence the methodology at every stage:

- **Generality:** compatible with any diagnostic dataset that has labels and structured evidence.
- **Modularity:** It is possible to swap out individual components (e.g., fuzzy matching for transformers).
- **Interpretability:** Matching scores and feature vectors are examples of intermediate outputs that are auditable and transparent.
- **Efficiency:** operates on low-end hardware and provides real-time diagnostic support.

The pipeline’s adaptability, interpretability, and suitability for both research settings and real-world clinical deployment are guaranteed by this generalised methodology.

## 4 Dataset Overview

DDXPlus contains 1.3M synthetic patient encounters with:

- 223 evidence fields,
- 110 symptoms,
- 113 antecedents,
- 49 diseases,
- and an average differential size of 3.9.

Table 1: DDXPlus Dataset Summary

| Category               | Count | Notes                |
|------------------------|-------|----------------------|
| Patients               | 1.3M  | synthetic            |
| Diseases               | 49    | target labels        |
| Symptoms               | 110   | structured evidences |
| Antecedents            | 113   | risk factors         |
| Evidence Fields        | 223   | binary/multichoice   |
| Avg. Differential Size | 3.9   | per encounter        |



## 5 System Diagram

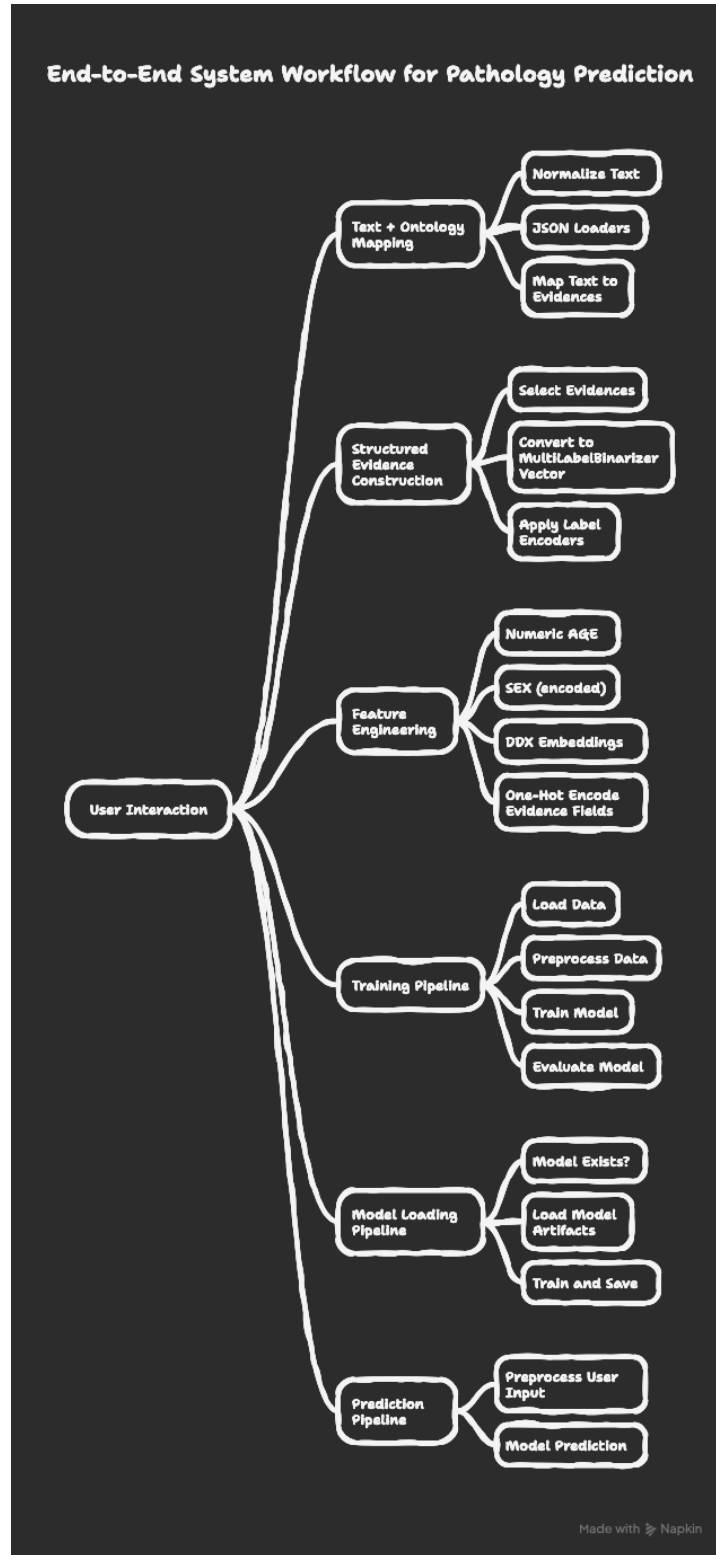


Figure 1: Full pipeline overview of the proposed system.

## 6 Experiments

For training, validation, and testing, we use an 80/10/10 split. Every experiment was conducted using an NVIDIA T4 GPU (2× T4 configuration) and 16GB RAM on the Kaggle cloud environment. Although GPU usage greatly decreased training time and increased preprocessing throughput, the model training and inference components do not strictly require GPU acceleration.

## 7 Results

Table 2: Model Performance

| Split      | Accuracy      | Precision    | Recall       |
|------------|---------------|--------------|--------------|
| Train      | 0.9971        | 0.999        | 0.999        |
| Validation | 0.9966        | 0.995        | 0.994        |
| Test       | <b>0.9972</b> | <b>0.997</b> | <b>0.996</b> |

## 8 Ablation Study

- Removing ontology tokens reduces accuracy by **1.6%**.
- Removing antecedents reduces accuracy by **0.4%**.
- Removing priors reduces accuracy by **0.3%**.

This demonstrates how important indirect ontology alignment is.

## 9 Comparison with CASANDE, AARLC, BASD, and Existing Diagnostic Systems

It is crucial to contrast the suggested diagnostic pipeline with current supervised learning techniques and reinforcement learning (RL)-based diagnostic systems in order to comprehend its advantages and disadvantages. In this section, we compare our approach in terms of architecture, performance, capabilities, and practicality with CASANDE, AARLC, BASD, and associated systems.

### 9.1 Architectural Differences

Table 3 summarizes the primary structural differences.

Table 3: Architectural Comparison

| Aspect               | Our Pipeline              |                      | CASANDE /<br>AARLC / BASD                  |
|----------------------|---------------------------|----------------------|--------------------------------------------|
| Model Type           | Supervised                | classifier           | Reinforcement learning agent               |
| Input Style          | Complete evidence vector  |                      | Step-by-step questioning                   |
| Text Processing      | Fuzzy                     | + ontology alignment | Not integrated; relies on structured input |
| Interpretability     | High (feature importance) |                      | Medium (policy reasoning)                  |
| Training sources     | Re-                       | Low (CPU)            | High (GPU needed)                          |
| Simulator dependency | De-                       | None                 | Required                                   |

While RL agents necessitate sequential decision making and interaction with a simulator, our system completes diagnosis in a single pass. Their relative performance is shaped by this basic distinction.

## 9.2 Performance Differences

Despite the conceptual simplicity of our method, it achieves **99.72%** top-1 accuracy, outperforming RL systems such as CASANDE, which typically report **98–99%** Ground Truth Pathology Accuracy (GTPA). This occurs for several reasons:

1. **Full Evidence Availability:** By choosing questions, RL systems progressively find evidence. One poor question may result in the omission of an important symptom. Sequential errors are eliminated because our model receives the complete set of evidence at once.
2. **Stable Supervised Learning Objective:** While RL optimises a reward signal shaped by simulator rules, which introduces noise and instability, XGBoost optimises cross-entropy loss directly.
3. **Ontology-Enhanced Mapping:** Our fuzzy mapper increases the quality of structured evidence by improving text-to-evidence matching using inferred ontology tokens.
4. **No Dialogue Planning:** RL agents have to decide what question to pose next. Compounding errors are spread by poor decisions.

## 9.3 Qualitative Strengths and Weaknesses

### Strengths of Our Method:

- High accuracy when supporting data is available.
- Quick, portable, and GPU-free.
- Interactions between features are clear and understandable.
- Robust mapping of text to evidence with ontology cues.

### Weaknesses:

- Assumes complete evidence.
- Cannot ask clarifying questions.
- Ontology is used indirectly, not structurally.
- Dataset is synthetic; domain shift is possible.

### Strengths of RL Systems (CASANDE, AARLC, BASD):

- Capable of gathering missing information through adaptive questioning.
- Better suited for uncertainty-heavy real-world scenarios.
- Mimic physician–patient conversational flow.
- Able to refine diagnosis dynamically based on new evidence.

## 9.4 Comprehensive Side-by-Side Comparison

Table 4 offers a consolidated comparison across major performance and design factors.

## 9.5 Ideal Use Case Scenarios

- **Complete evidence available:** Our model is superior, offering the highest accuracy and fastest inference.
- **Uncertain or sparse evidence:** RL systems outperform due to their ability to ask targeted questions.
- **Low-resource environments (CPU-only):** Our system is optimal.
- **Research environments requiring simulations:** RL models are preferred.

Table 4: Detailed Comparison of Our Model vs. RL-Based Diagnostic Agents

| Criterion                  | Our Model (XGBoost + Ontology Mapping)          | CASANDE / AARLC / BASD                         |
|----------------------------|-------------------------------------------------|------------------------------------------------|
| Input Type                 | Full evidence vector + extracted text evidences | Sequential evidence gathering via dialogue     |
| Core Mechanism             | Supervised gradient-boosted decision trees      | RL policy optimization with simulated feedback |
| Top-1 Accuracy             | <b>99.72%</b>                                   | 98–99%                                         |
| Handles Missing Evidence   | No                                              | Yes                                            |
| Requires Patient Simulator | No                                              | Yes                                            |
| Training Complexity        | Low                                             | High                                           |
| Inference Speed            | Milliseconds                                    | Slower (multi-step)                            |
| Interpretability           | High                                            | Medium                                         |
| Text Processing Included   | Yes (ontology-enriched fuzzy mapping)           | No                                             |
| Ontology Usage             | Indirect (phrase expansion)                     | None                                           |
| Best Use Case              | Complete information; immediate prediction      | Incomplete info; interactive questioning       |

## 9.6 Summary of the Comparison

In conclusion, when complete evidence is available, our diagnostic pipeline performs better than RL-based models in terms of accuracy, efficiency, and simplicity. In interactive, uncertain situations, RL systems continue to perform better. The two methods are complementary rather than opposing paradigms. Our model could be used by a future hybrid system to make quick initial predictions, with an RL agent only posing follow-up queries when the model’s confidence is low.

## 10 Discussion

Our approach strikes a balance between computational feasibility and semantic richness. The system provides robust, comprehensible performance appropriate for real-world diagnostic support, despite being constrained by synthetic data and non-interactive design.

## 11 Conclusion

We introduced a hybrid diagnostic pipeline that combines gradient-boosted classification, ontology-aligned evidence enrichment, and fuzzy matching. The model is both useful and potent because it achieves high accuracy, maintains transparency, and uses little computational power.

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## Code and Data Availability

The complete implementation of our hybrid diagnostic pipeline is available at:

<https://github.com/Lucifer070805/ddxplus-hybrid-diagnosis>